

# NE 511: Multi-Physics of Nuclear Reactors

## UNIT 2: Short-Time Multi-Physics Phenomena in Reactor Core

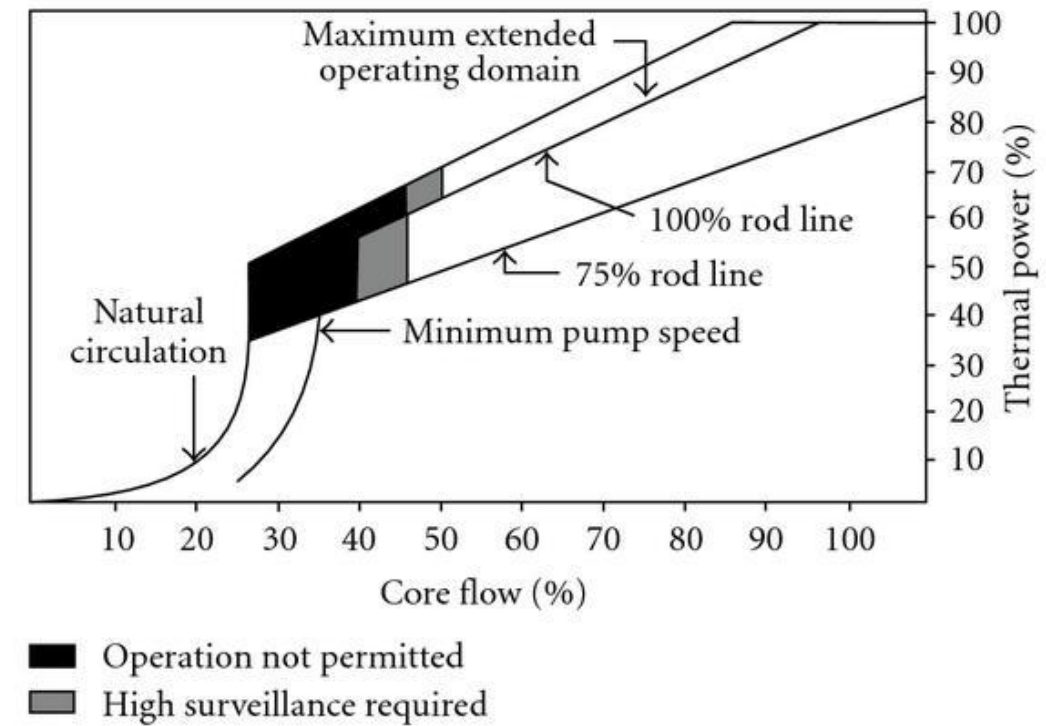
### Lecture 12: BWR INSTABILITIES

## BWR INSTABILITIES

At a relatively high power-to-flow ratio, BWR plants are prone to instabilities → Reactor operation is restricted.

*Needed Action:* SCRAM.

*Main consequences:* Economic losses due to unavailability of the plant.



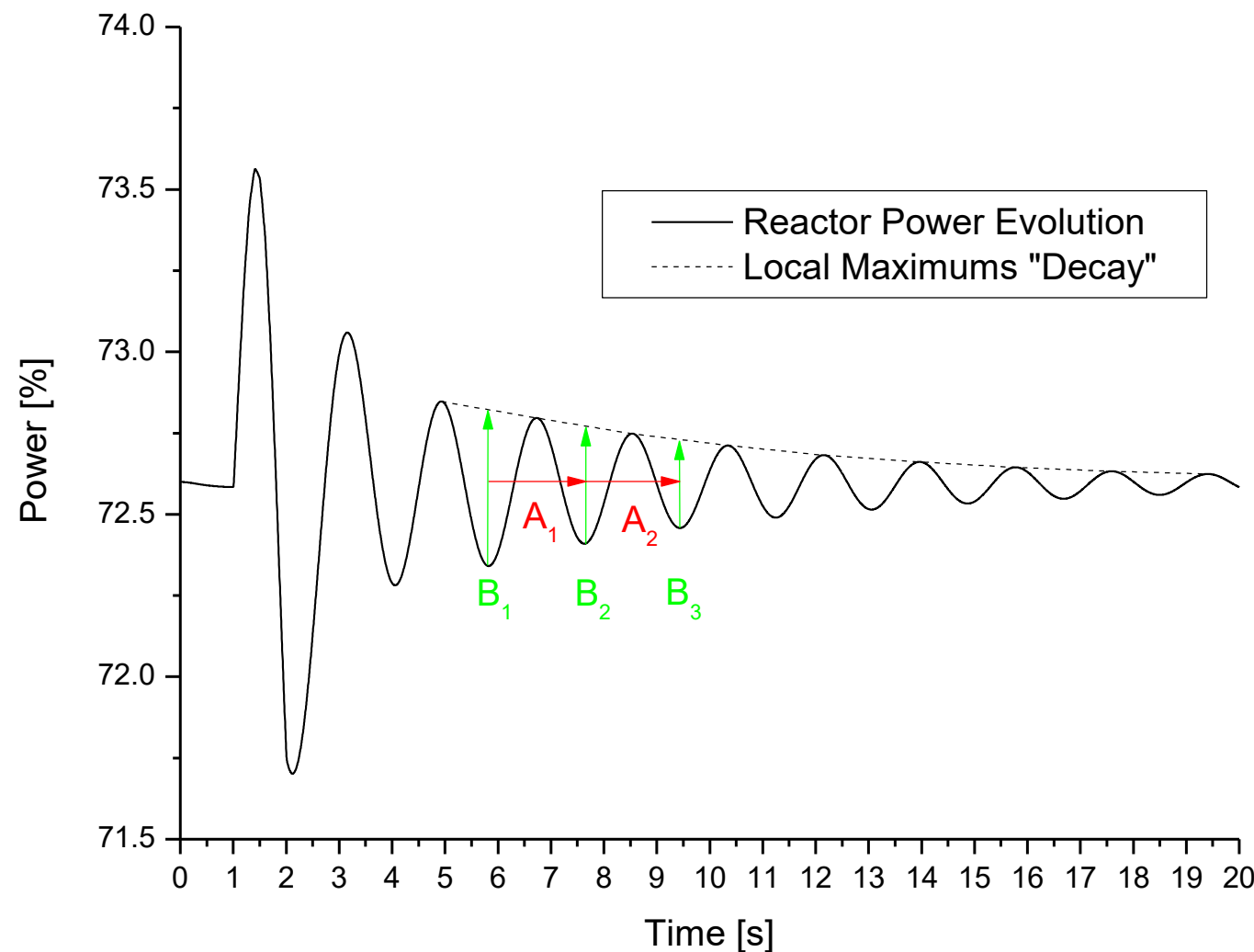
## SAFETY ISSUE

- !! Undetected local oscillations, may bring local power beyond licensing limits – causing unacceptable rod surface temperature.

*In any case, the reliability of SCRAM and the efficiency of instrumentation, ensure the BWR stability safety.*

## HOW TO QUANTIFY THE BWR STABILITY?

1. Decay Ratio is the relative ratio of two consecutive oscillations.
2. Frequency is the frequency of oscillation of the reactor power signal.



$$DR = \frac{1}{2} \left( \frac{B_2}{B_1} + \frac{B_3}{B_2} \right)$$

$$FR = \frac{2}{A_1 + A_2} [Hz]$$

## BWR INSTABILITIES

The BWR instabilities occurred in operating NPP may be distinguished in the following categories:

1. Fundamental mode is the in-phase (or global) mode, where all core regions oscillate together in-phase;
2. The first harmonic mode is the out-of-phase (or regional) mode, where one half of the core oscillates out of phase against the other half;
3. Local (or channel) oscillations: only one channel or a few channels experience density wave propagation throughout the entire channel.

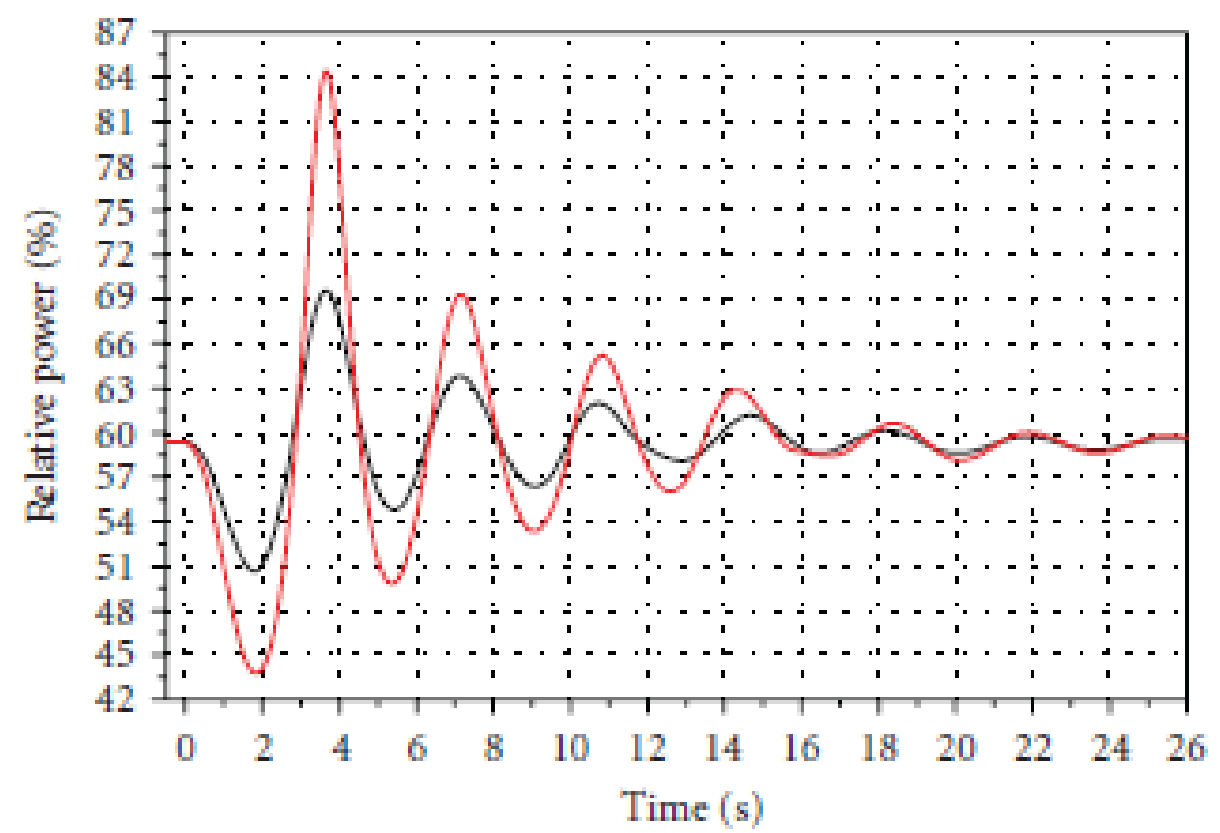
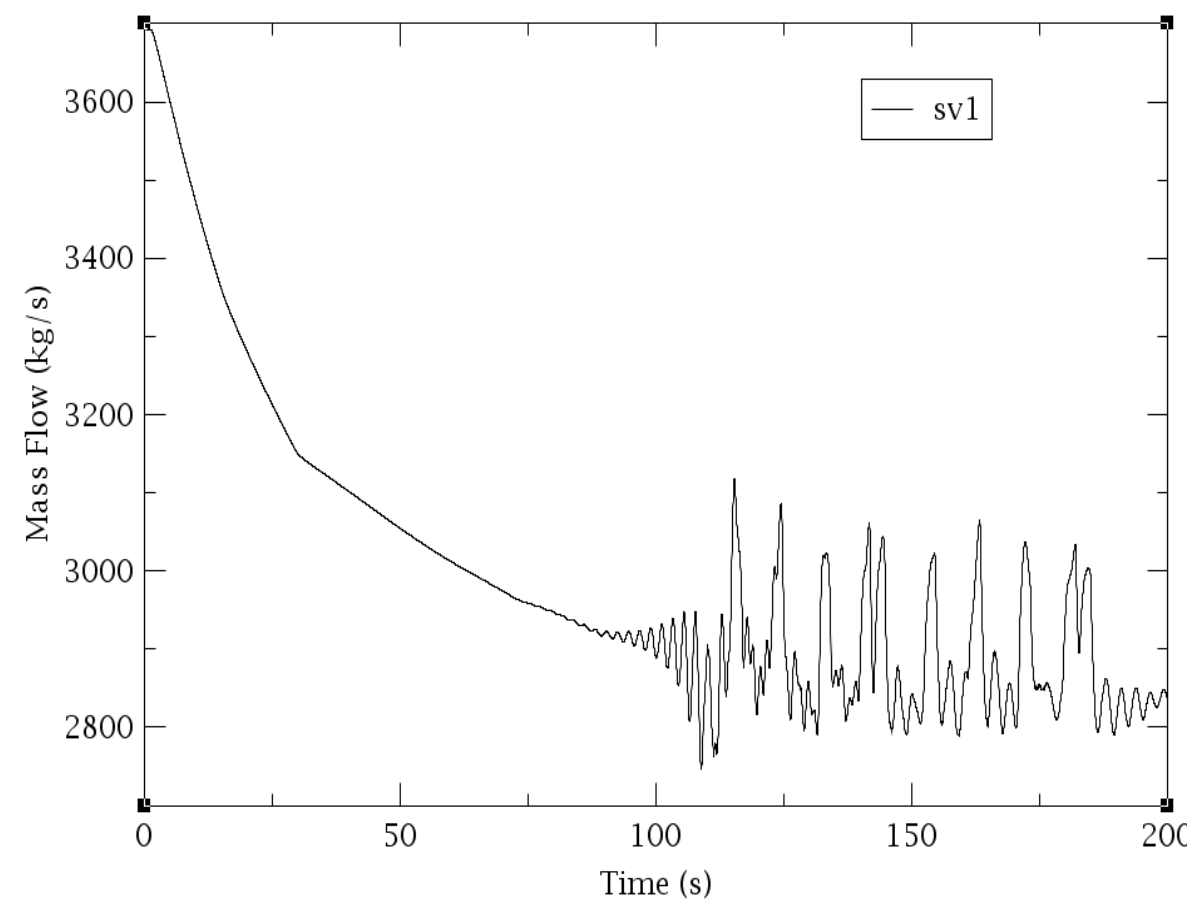
# BWR INSTABILITIES EXAMPLE - RINGHALS-1 REACTOR

## TRACE/PARCS Simulations

### Recirculation pump trip

All recirculation pumps lost their torques at 1 second.

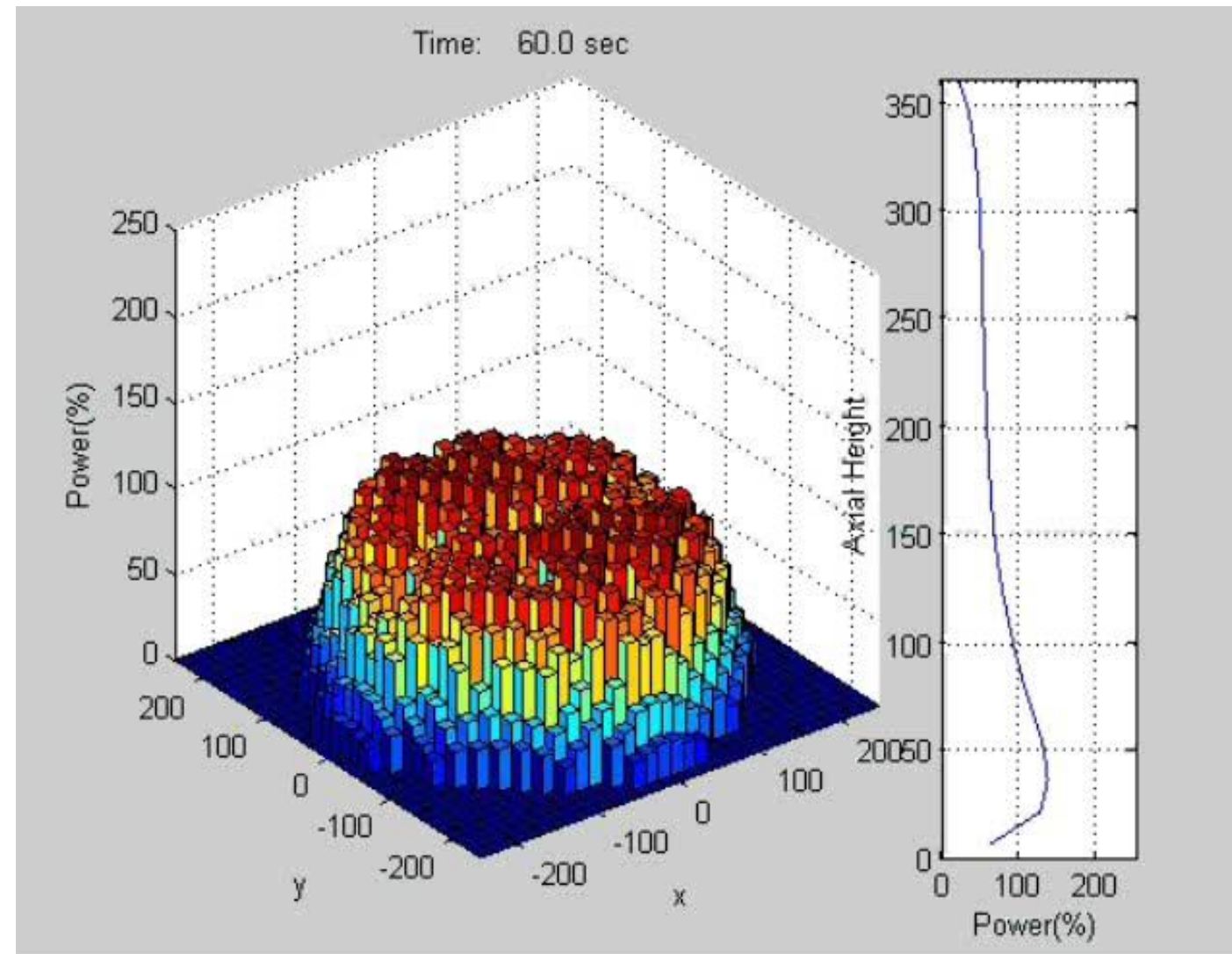
Feed water temperature coast down to 350 Kelvin ( $\sim 170$  F /  $\sim 77$  °C) in 60 seconds.



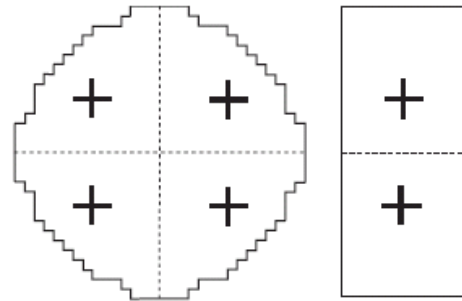
# BWR INSTABILITIES EXAMPLE - RINGHALS-1 REACTOR

*TRACE/PARCS Simulations*

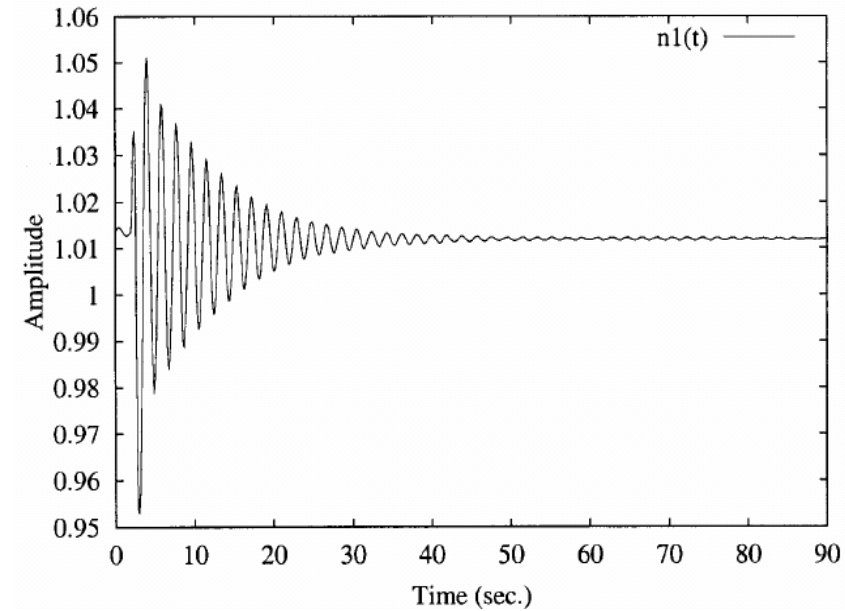
**Recirculation pump trip - Power response**



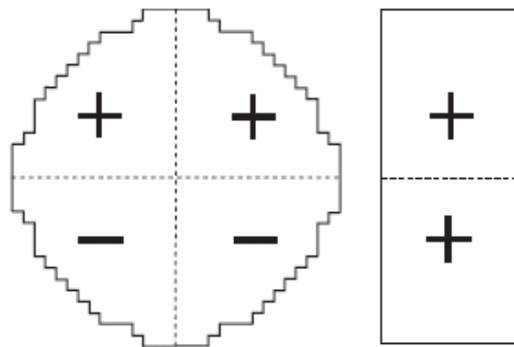
# BWR INSTABILITIES EXAMPLE - RINGHALS-1 REACTOR



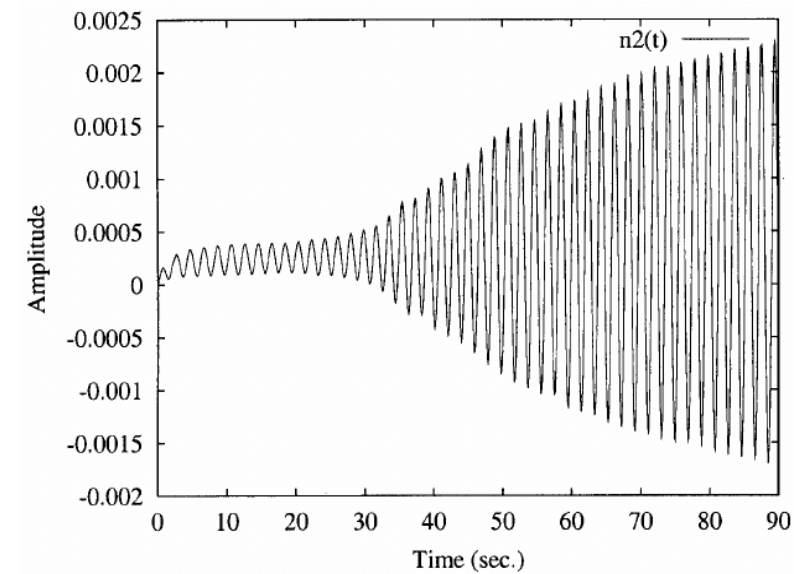
Power profile for the first mode  
(fundamental mode, radial)



Evolution of the amplitude associated with the  
fundamental azimuthal mode

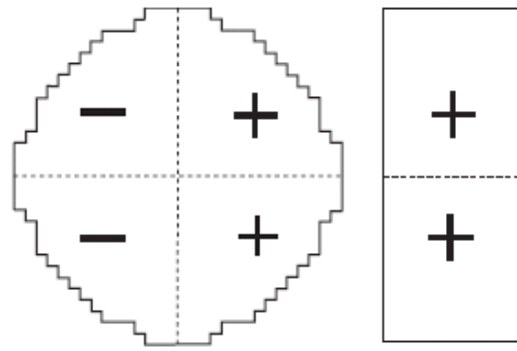


Power profile for the second mode  
(first azimuthal mode)

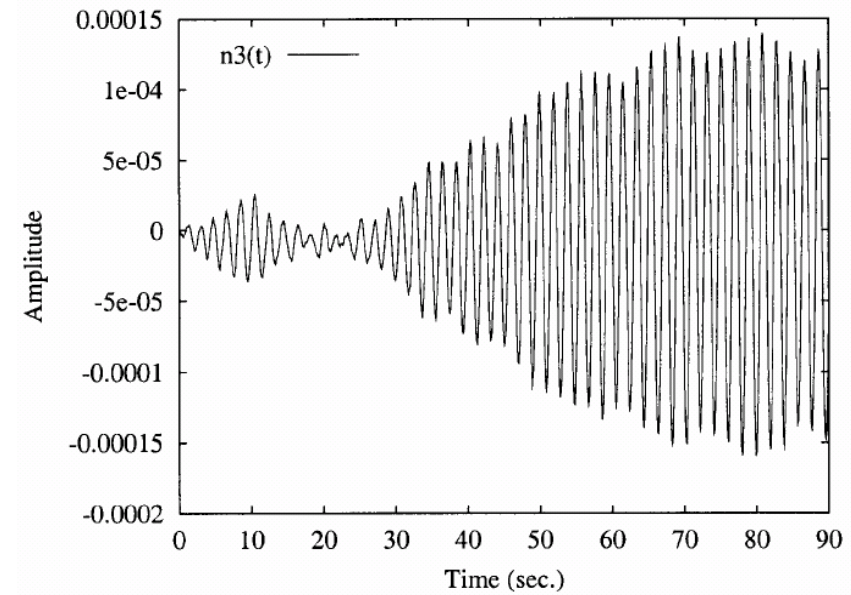


Evolution of the amplitude associated with the first  
azimuthal mode

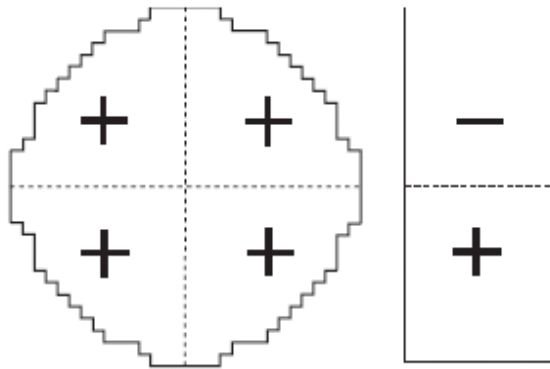
# BWR INSTABILITIES EXAMPLE - RINGHALS-1 REACTOR



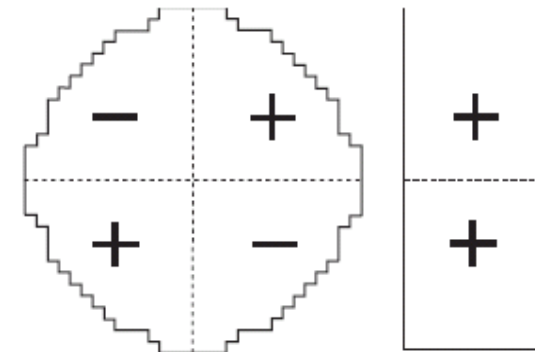
Power profile for the third mode  
(second azimuthal mode)



Evolution of the amplitude associated with the  
second azimuthal mode



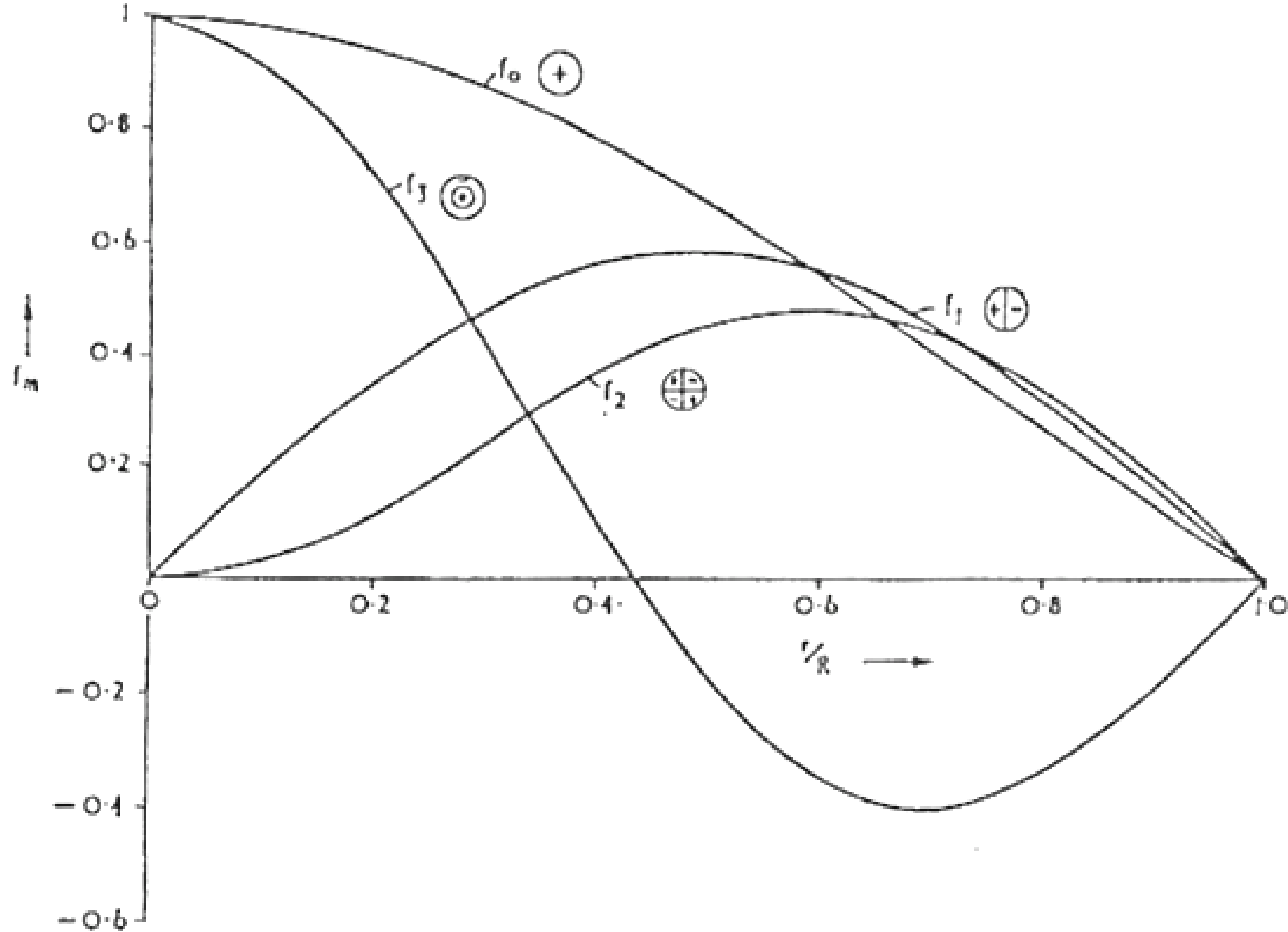
Power profile for the fourth mode  
(axial mode)



Power profile for the fifth mode (third  
azimuthal mode)



# BWR INSTABILITIES EXAMPLE - RINGHALS-1 REACTOR



# BENCHMARKS ON BWR STABILITY



- Ringhals-1
  - Stability tests at BOC14, BOC15, BOC16, MOC16, BOC17.
  - Time domain and frequency domain solution is possible.
- Forsmark-1 and -2
  - Measured APRM and LPRM data.
  - Analysis of time series data by noise analysis techniques in the time domain.
- **BWR stability benchmark**
  - **Based on an actual stability event – Oskarshamn-2 event on February 25, 1999.**
  - **Swedish type 1800 MW external loop BWR.**
  - **Challenging to neutronics, thermal-hydraulics, coupling.**

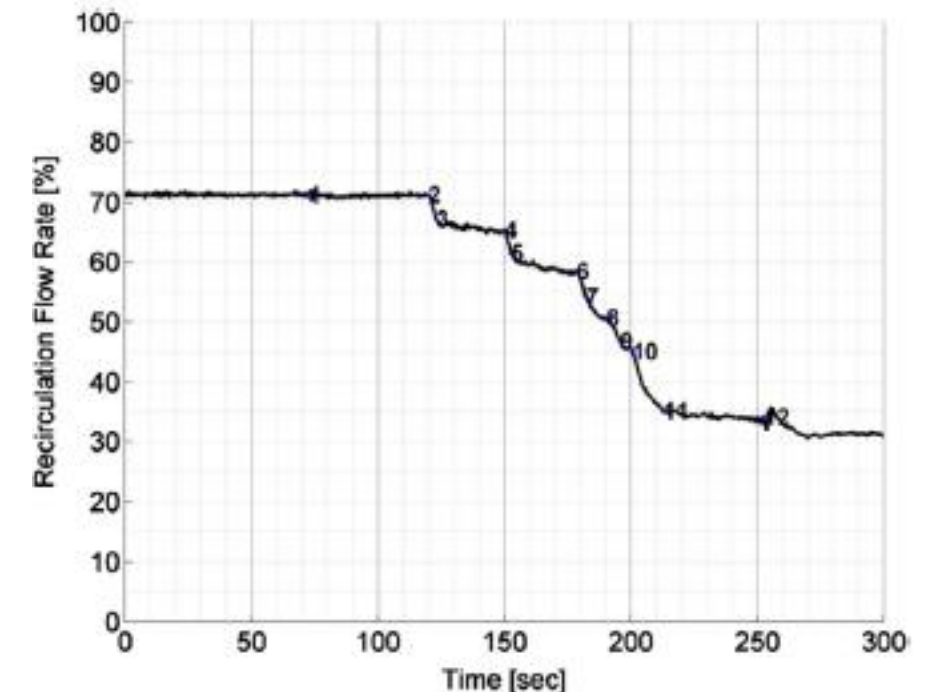
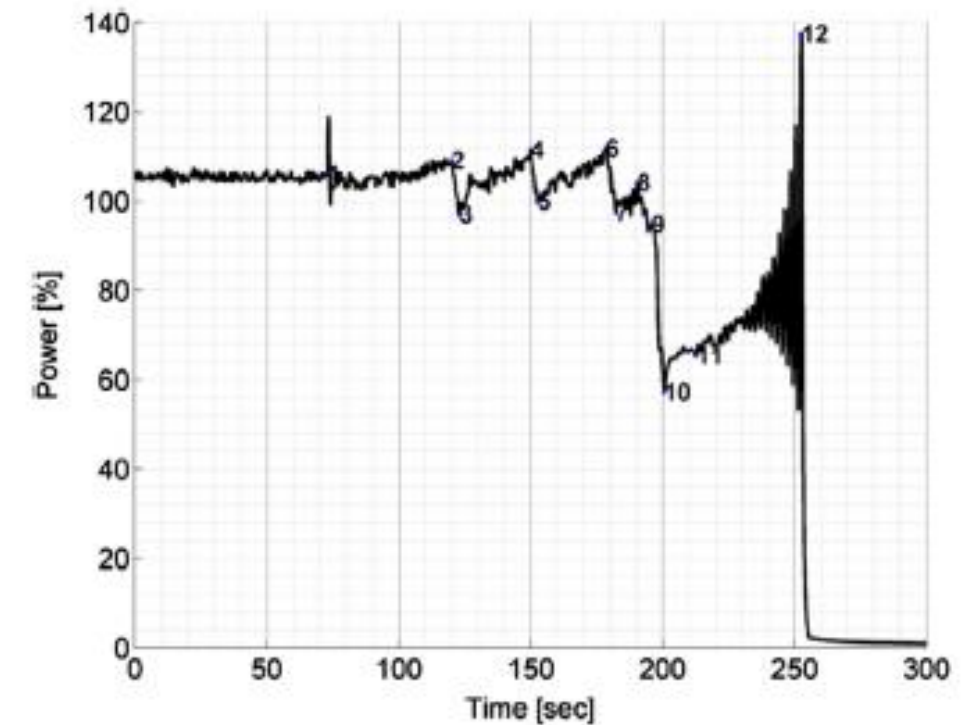
## OSKARSHAMN-2 1999 STABILITY EVENT - DESCRIPTION

Loss of feed-water preheaters and regular operation of automatic power/flow control system caused the plant to move into low flow – high power regime which triggered power oscillations:

- *A load rejection signal was transmitted to the turbine and caused a turbine trip.*
- *The signal was not transferred to the reactor and it kept on running at nominal power.*
- *Because of the turbine trip, the feedwater preheaters tripped and the feedwater temperature started decreasing.*
- *Colder feedwater temperature created a positive reactivity feedback, increasing the core power level.*
- *Automatic system reduced the main recirculation flow, thereby reducing the power.*
- *This sequence was repeated three times at 100, 125, and 150 seconds.*

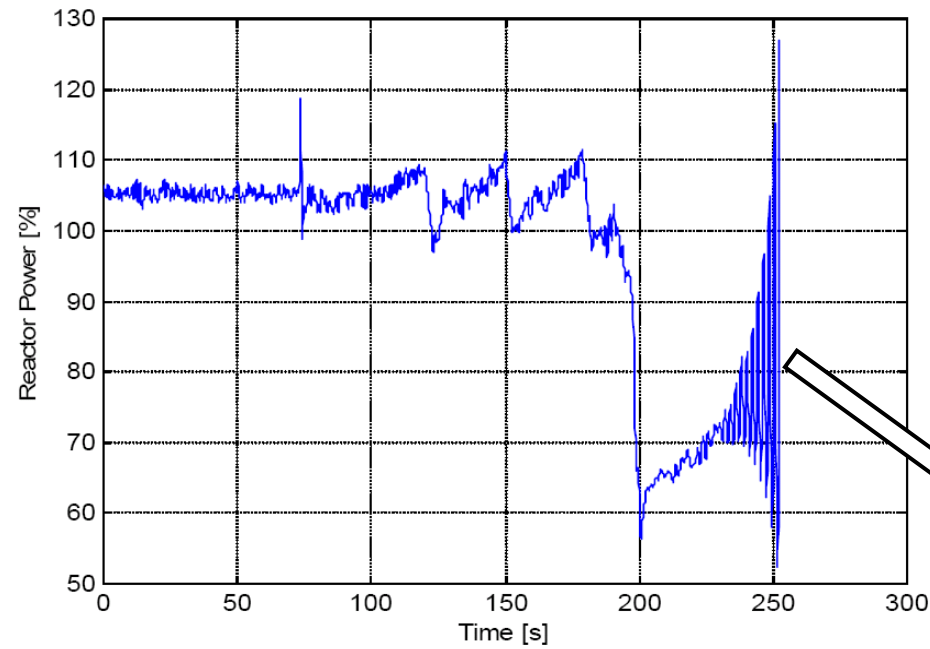
## OSKARSHAMN-2 1999 STABILITY EVENT - DESCRIPTION

- Short loss of external power leads to turbine trip. Closure of turbine control valves and opening of the dump valves (1)
- Generator power reduced, but no pump run-down – reactor remains at full power.
- Loss of FW pre-heaters, FW flow remained constant.
- **Reduced FW temperature** causes decrease in coolant temperature which causes power increase until 108% (2)
- **Recirculation flow reduced** to decrease power. This sequence was repeated two more times (4, 6)
- Operators trigger **partial scram**, recirculation pump run-down to minimum pump speed (8, 9). Power decreased to 65% (10).
- Continuing decrease of FW and coolant temperature **pushes reactor into unstable conditions (11)**.
- Reactor scrammed at high power (132%) after 20 s (12).

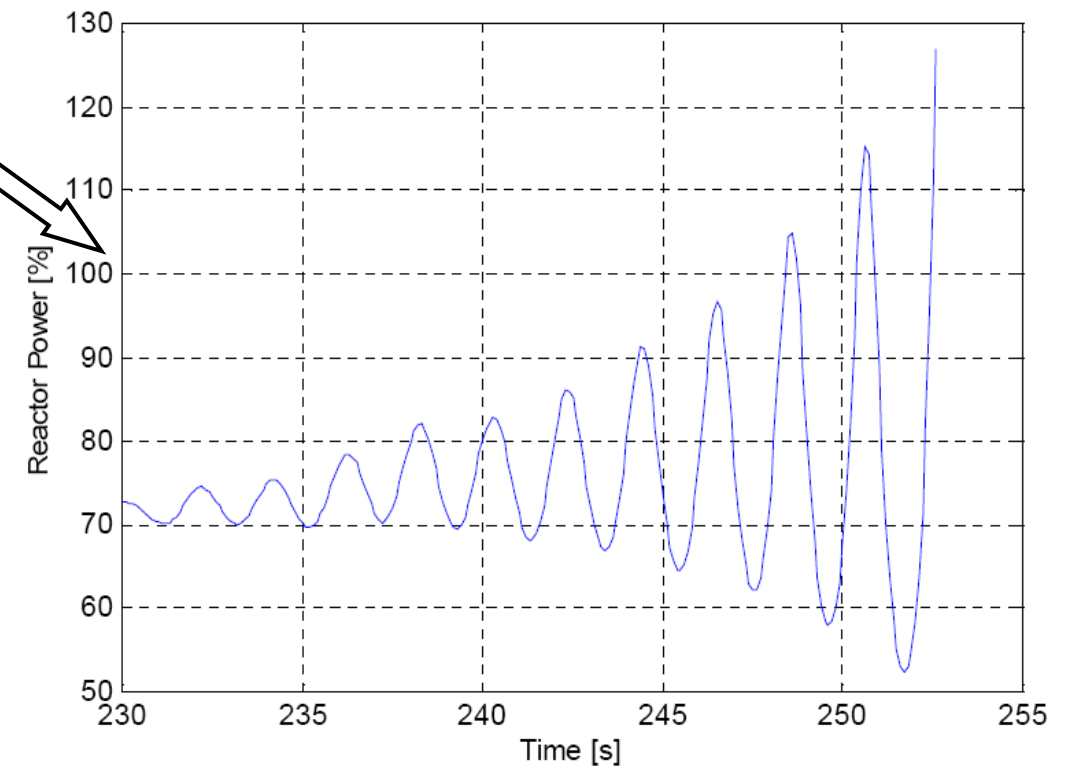


# OSKARSHAMN-2 1999 STABILITY EVENT - DESCRIPTION

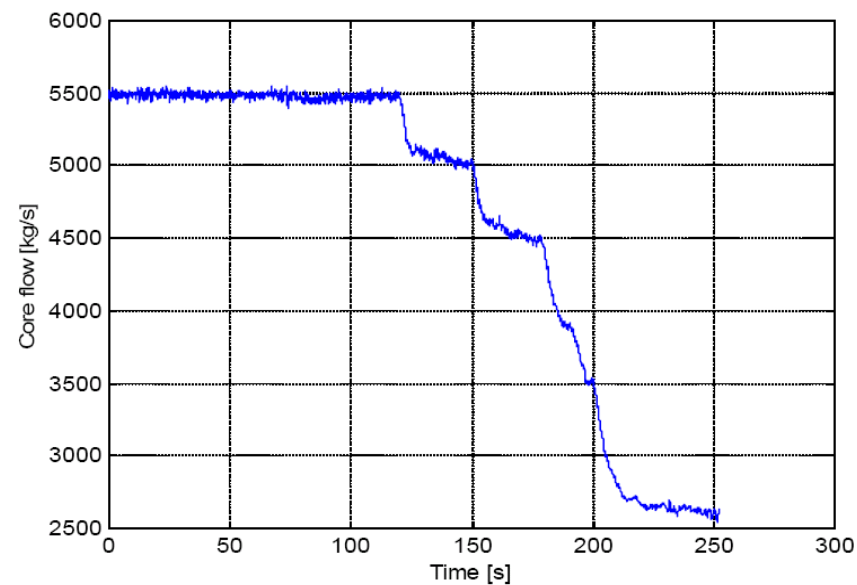
Feedwater Transient: *Real Plant Data*



Feedwater Transient: *Real Plant Data*



Feedwater Transient: *Real Plant Data*

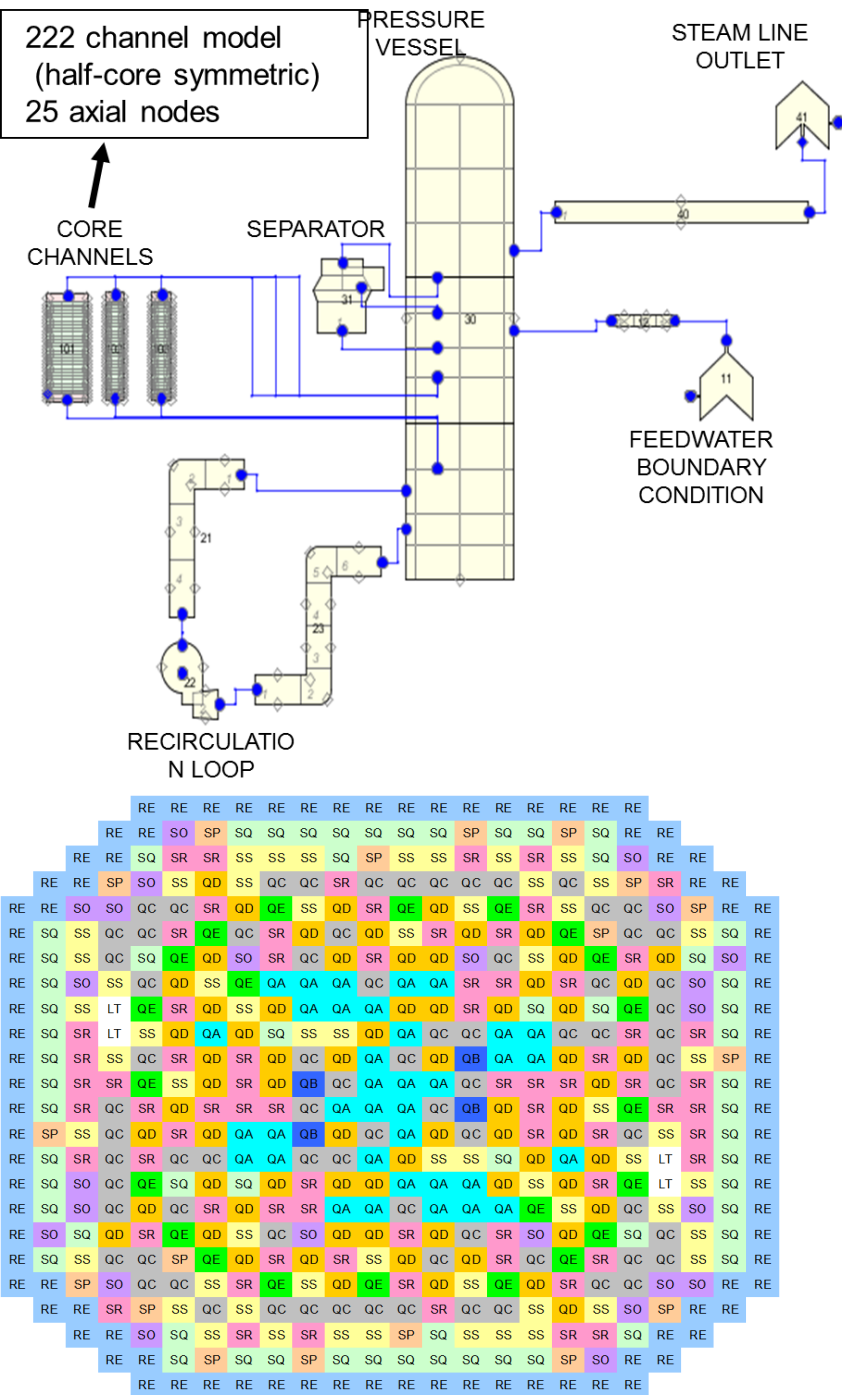


# OSKARSHAMN-2 TRACE/PARCS MODEL

For steady-state convergence, the following controllers are used:

| BOP Controllers:        | Component Being Adjusted:  |
|-------------------------|----------------------------|
| Recirculation flow rate | Recirculation pump speed   |
| Steam dome pressure     | Steam line outlet pressure |
| Downcomer water level   | Feedwater flow rate        |
| Core inlet subcooling   | Feedwater temperature      |

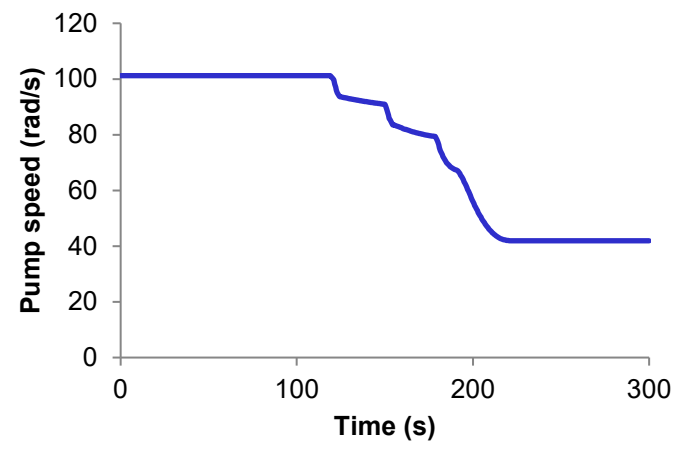
→ For the transient calculation, the BOP controllers are replaced by time-dependent boundary conditions.



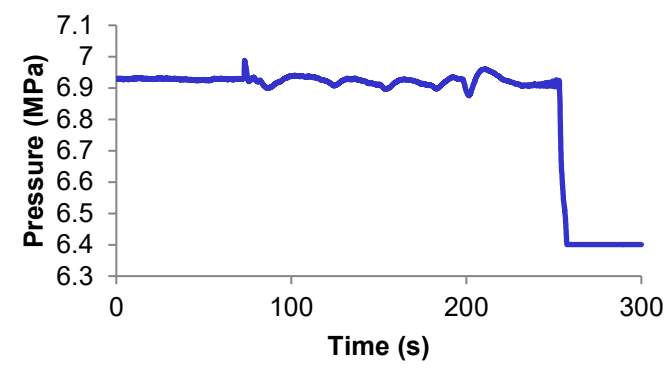
# OSKARSHAMN-2 TRACE/PARCS MODEL

## *Boundary Conditions & Transient Results*

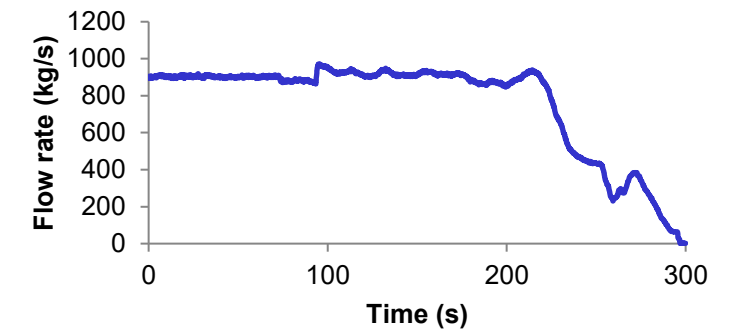
**Pump Speed**



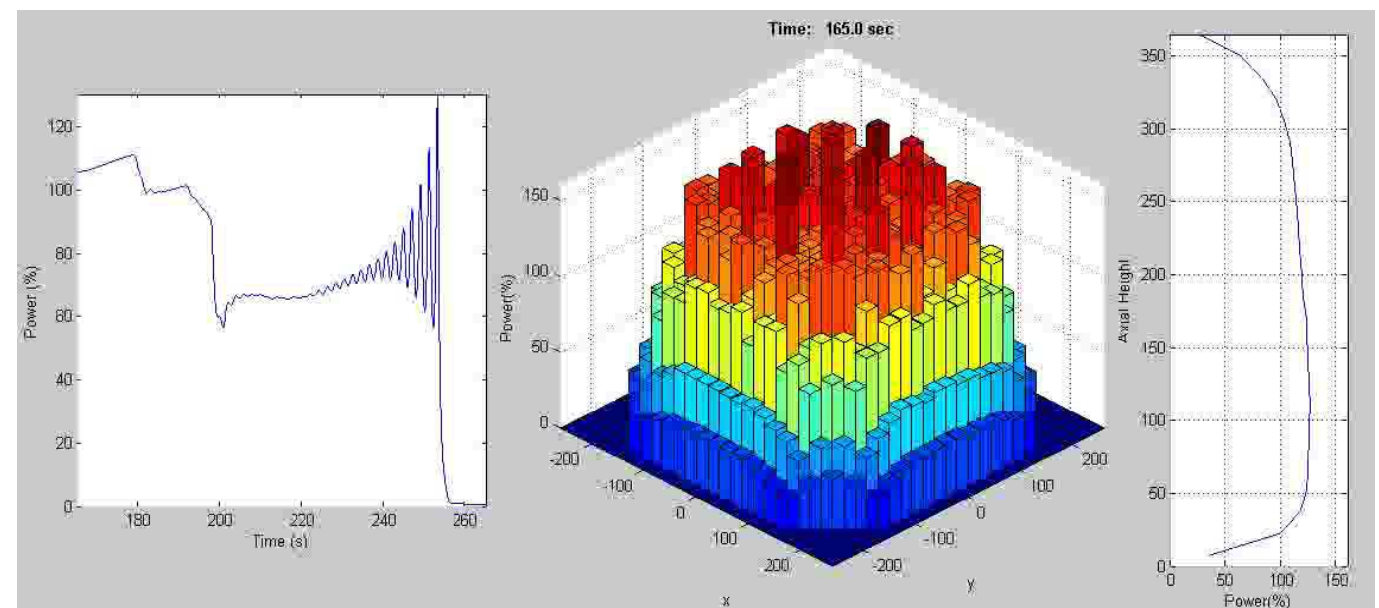
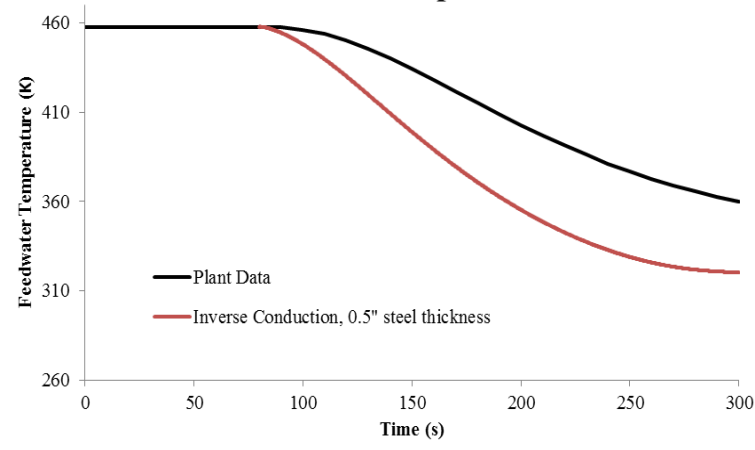
**Steam Dome Pressure**



**Feedwater Flow Rate**



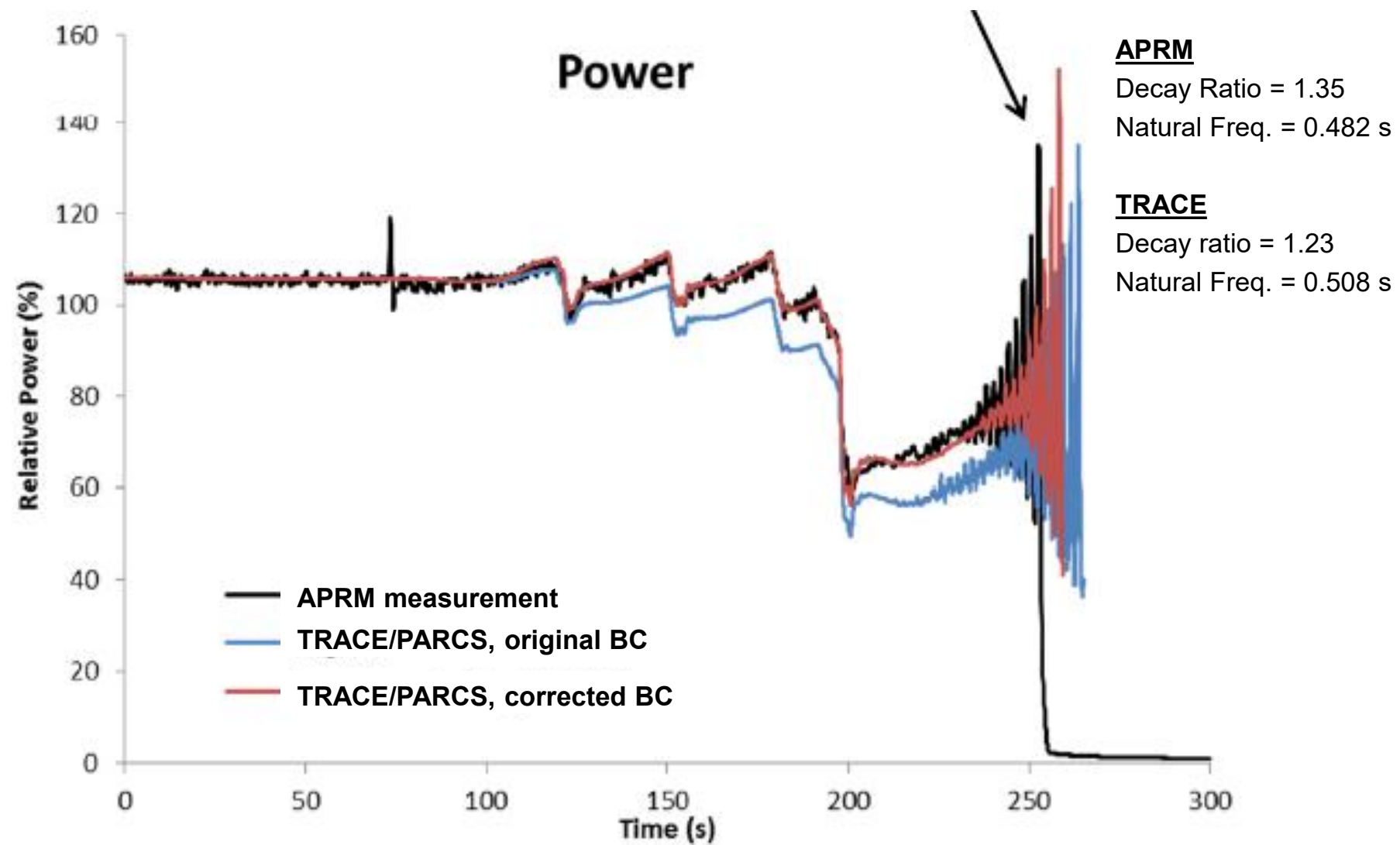
**Feedwater Temperature**





# OSKARSHAMN-2 TRACE/PARCS MODEL

## *Transient Results*





# GENERAL REACTOR STABILITY

## GENERAL REACTOR STABILITY

After combining all the feedback effects, the following set of power reactor equations is obtained:

$$l \left[ \frac{\partial \phi}{\partial t} + \sum_i \frac{\partial \phi_i}{\partial t} \right] = M^2 \nabla^2 \phi + (k - 1) \phi + \delta k \phi$$

$$l \frac{\partial \phi_i}{\partial t} = k \beta_i \phi - \lambda_i l \phi_i$$

$$k = k^* + \alpha_f \Delta T_f + \alpha_m \Delta T_m + \alpha_X \Delta X + \dots$$

$$\frac{\partial I}{\partial t} = s \sigma_X \phi - \lambda_I I$$

$$\frac{\partial X}{\partial t} = \lambda_I I + (1 - s) \sigma_X \phi - \lambda_X X - \sigma_X X \phi$$

$l$ : neutron lifetime

$$M^2 = \frac{D}{\Sigma_a}$$

$$\phi_i = v p_i N_i$$

together with suitable auxiliary equations describing,  $T_f$ ,  $T_m$ ,  $Xe$ , ... in terms of  $f$ .

*A set of coupled, non-linear differential equations for given initial conditions and disturbances.*

# GENERAL REACTOR STABILITY

This set of coupled, non-linear, differential equations presents a very difficult problem - an analytical (or near-analytical) treatment seems hardly feasible.

Some attempt has been made to study the stability of rather simpler sets of non-linear equations:

1. *Consider small perturbations to an equilibrium steady-state solution and investigate in what circumstances the perturbations will grow or subside.*
2. *Consider only small disturbances and enable an effective linearization of the relevant equations - products of small quantities may be neglected.*

# GENERAL REACTOR STABILITY

Consider that the time-dependent quantities receive small perturbations from their steady-state values, i.e.

$$\phi = \phi^* + \phi'$$

Then neglecting the product of small (primed) quantities we have ...

$$l \left[ \frac{\partial \phi'}{\partial t} + \sum_i \frac{\partial \phi'_i}{\partial t} \right] = M^2 \nabla^2 \phi' + (\alpha_f T_f' + \alpha_m T_m' + \dots) \phi^* + (k^* - 1) \phi' + \delta k (\phi^* + \phi'),$$

$$l \frac{\partial \phi'_i}{\partial t} = (\alpha_f T_f' + \alpha_m T_m' + \dots) \beta_i \phi^* + k^* \beta_i \phi'_i - \lambda_i l \phi'_i$$

$$\frac{\partial I'}{\partial t} = s \sigma_X \phi' - \lambda_I I' \quad \frac{\partial X'}{\partial t} = \lambda_I I' + (1 - s) \sigma_X \phi' - \lambda_X X' - \sigma_X X^* \phi' - \sigma_X \phi^* X'$$

## GENERAL REACTOR STABILITY

... together with heat-transfer equations for fuel, moderator and coolant

$$\frac{\partial T_f'}{\partial t} = K_f \phi' - (T_f' - T_c') \gamma_f$$

$$\frac{\partial T_m'}{\partial t} = K_m \phi' - (T_m' - T_c') \gamma_m$$

$$V \frac{\partial T_c'}{\partial z} = (T_f - T_c) \gamma_{fc} + (T_m' - T_c') \gamma_{mc}$$

To investigate the time behavior of this set of equations, the flux perturbations can be expressed as a sum of terms, each separable in time and space:

$$\phi' = \sum_m A_m a_m(t) h_m(r, \theta, Z)$$

## GENERAL REACTOR STABILITY

Ideally, the  $\underline{h}_m$  functions are chosen to be the normal modes of the set of equations (in the same way as the simple neutronics equations without feedback are treated).

The time dependence associated with these modes has the form:

$$a_m(t) = e^{\omega_m t}$$

*The values of  $\omega_m$  are determined only by the equations themselves and not by the nature of the initial disturbance.*

*The initial conditions, however, determine the values of  $\alpha_m$  - i.e. the magnitudes of the modal components.*

*Positive omega means that an initial disturbance will keep increasing leading to an unstable system.*

*The objective is to discover in what circumstances any particular value of  $\omega_m$  is positive (or has positive real part).*

# GENERAL REACTOR STABILITY

To recall:

The power change due to the temperature change is:

$$\delta P = -\frac{\delta k}{(\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^*)} \left\{ 1 + \frac{\alpha_m \Delta T_m^*}{\alpha_f \Delta T_f^*} e^{-\omega t} \right\}$$

where  $\omega = \frac{(\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^*)}{\alpha_f \Delta T_f^* \tau_m}$

Since  $\alpha_f \Delta T_f^*$  is always negative in practice, the instability criterion may also be written

$$\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^* > 0$$



# GENERAL REACTOR STABILITY

The harmonics are given by the eigenvalue equation

$$\nabla^2 F_m + \frac{(k^* - 1)}{M^2} F_m + \mu_m F_m = 0$$

- ▶ The fundamental ( $m=0$ ) harmonic is associated with the eigenvalue  $m_0 = 0$  and is clearly proportional to the steady-state flux level.
- ▶ The higher harmonics are related to the normal modes of the neutronics equation with no feedback.
- ▶ Thus, when the feedback is reasonably small, we may anticipate that the chosen harmonics are fairly close approximations to the actual modes of the reactor system.

## GENERAL REACTOR STABILITY

- ▶ The magnitudes of cross-coupling terms (between the chosen harmonics) are found to be small - cross-coupling terms can be ignored.
- ▶ For a cylindrical reactor with a constant value  $(k^*-1)/M^2$  the harmonics are given simply by:

$$F_m \equiv F_{l\gamma n} = A_m J_\gamma(\beta_{\gamma l} r) \cos \gamma \theta \sin \frac{n\pi z}{L}$$

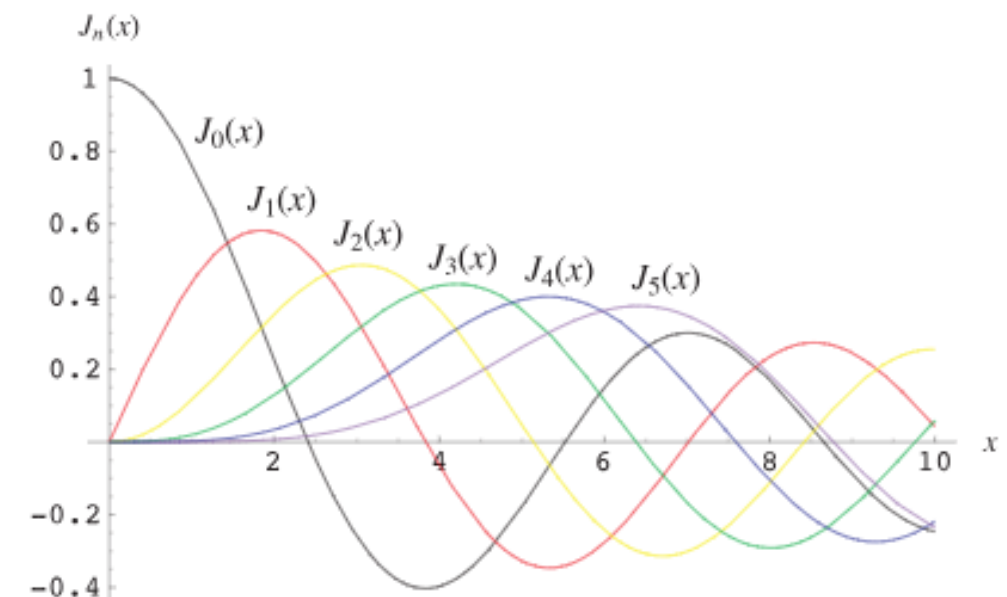
where  $J_g$  denotes a Bessel function of order  $g$  and the quantities  $l, g, n$  take integer values only

***The suffix  $l$  indicates the particular zero of the Bessel function considered and  $L$  is the reactor (extrapolated) height.***

Fundamental mode:  $m = 0 \rightarrow l = 1, \gamma = 0, n = 1$

First azimuthal mode:  $m = 1 \rightarrow l = 1, \gamma = 1, n = 1$

Second azimuthal mode:  $m = 1 \rightarrow l = 1, \gamma = 2, n = 1$



## GENERAL REACTOR STABILITY

The eigenvalues are given by:

$$\frac{n^2 \pi^2}{L^2} + \beta_{\gamma l}^2 = \frac{k^* - 1}{M^2} + \mu_m$$

The fundamental harmonic ( $m = 0$ ) is given by:

$$l = 1; \quad \gamma = 0; \quad n = 1$$

The first azimuthal harmonic ( $l = 1; g = 1; n = 1$ ) will be denoted by  $m = 1$ , the second azimuthal harmonic ( $l = 2; g = 2; n = 2$ ) by  $m = 2$ , and so on.

## GENERAL REACTOR STABILITY

In general, the radial azimuthal harmonics of interest will be written in the following form:

$$F_m \equiv f_m(r, \theta) g_0(z)$$

where  $g_0(z) = \sin(\pi z/L)$  is the steady-state axial flux form.

Using the condition that  $f_m$  should vanish at the reactor boundary we have from the equation of the eigenvalues and the properties of Bessel functions:

$$\mu_0 R_0^2 = 0, \quad \mu_1 R_0^2 = 8.90, \quad \mu_2 R_0^2 = 20.59, \text{etc.},$$

where  $R_0$  is the (extrapolated) reactor radius.

# GENERAL REACTOR STABILITY

To the expansion of the flux perturbation, we may write:

$$\phi' = \sum_m A_m e^{\omega_m t} f_m(r, \theta) g_0(z)$$

Further, since the delayed neutrons,  $^{135}\text{Xe}$  and  $^{135}\text{I}$  distributions are local functions of the neutron flux, we may write:

$$\phi_l' = \sum_m B_{ml} e^{\omega_m t} f_m(r, \theta) g_0(z)$$

$$i' = \sum_m C_m e^{\omega_m t} f_m(r, \theta) g_0(z)$$

$$x' = \sum_m D_m e^{\omega_m t} f_m(r, \theta) g_0(z)$$

# GENERAL REACTOR STABILITY

Now:

1. Substitute the last equations into the heat transfer equations for fuel, moderator, and coolant:

$$\begin{array}{lcl}
 \phi' = \sum_m A_m e^{\omega_m t} f_m(r, \theta) g_0(z) & & \frac{\partial T_f'}{\partial t} = K_f \phi' - (T_f' - T_c') \gamma_f \\
 \phi_l' = \sum_m B_{ml} e^{\omega_m t} f_m(r, \theta) g_0(z) & \longrightarrow & \frac{\partial T_m'}{\partial t} = K_m \phi' - (T_m' - T_c') \gamma_m \\
 i' = \sum_m C_m e^{\omega_m t} f_m(r, \theta) g_0(z) & & V \frac{\partial T_c'}{\partial z} = (T_f - T_c) \gamma_{fc} + (T_m' - T_c') \gamma_{mc} \\
 x' = \sum_m D_m e^{\omega_m t} f_m(r, \theta) g_0(z) & & 
 \end{array}$$

2. Multiply by  $F_n$ ;
3. Integrate over the reactor core; and  
*(Since we are concerned only with the  $\omega_m$  we omit the applied multiplication  $\delta k$ )*
4. Neglect cross-coupling between the harmonics  $\rightarrow$  (next slide)

## GENERAL REACTOR STABILITY

Thus,

$$l\omega_n + k_n^* \omega_n \sum_i \frac{\beta_i}{\omega_n + \lambda_i} - \alpha_{Xk} \phi_n^* \Gamma_n + M^2 \mu_n - \int_V (\alpha_f T_n' + \alpha_m T_m') \phi^* f_n g_0 dV = 0$$

where

$$\Gamma_n = \frac{\sigma_X (1-s) - x_n^* + \frac{\lambda_I s}{\omega_n + \lambda_I}}{\omega_n + \lambda_X + \sigma_X \phi_n^*}$$

is the Xenon transfer function and we have included only  $^{135}\text{Xe}$  of the fission product poisons.

# GENERAL REACTOR STABILITY

- The evaluation of the integral occurring in the last equation requires a consideration of the heat-transfer equation.

Before doing that, we make one further assumption - that the heat required to change the temperature of the various components is negligible in comparison with the heat removed from the reactor, i.e.

$$C_f \frac{\partial T_f}{\partial t} + C_m \frac{\partial T_m}{\partial t} + C_c \frac{\partial T_c}{\partial t} \ll V C_c \frac{\partial T_c}{\partial z}$$



$$\int_V T_f' \phi^* f_n g_0 dV = \frac{\gamma_f T_{fn}^*}{\omega_n + \gamma_f} \quad \text{and} \quad \int_V T_m' \phi^* f_n g_0 dV = \frac{\gamma_m T_{mu}^*}{\omega_n + \gamma_m}$$

where  $g_f = 1/t_f$ , etc.



# GENERAL REACTOR STABILITY

$$l\omega_n + k_n^* \omega_n \sum_i \frac{\beta_i}{\omega_n + \lambda_i} - \alpha_X \phi_n^* \Gamma_n + M^2 \mu_n - \int_V (\alpha_f T_n' + \alpha_m T_m') \phi^* f_n g_0 dV = 0$$

$$\int_V T_f' \phi^* f_n g_0 dV = \frac{\gamma_f T_{fn}^*}{\omega_n + \gamma_f}$$

$$\int_V T_m' \phi^* f_n g_0 dV = \frac{\gamma_m T_{mu}^*}{\omega_n + \gamma_m}$$

$$H_n(\omega_n) \equiv l\omega + k_n^* \omega \sum_i \frac{\beta_i}{\omega + \lambda_i} - \alpha_X \phi_n^* \Gamma_n + M^2 \mu_n - \frac{\gamma_f T_{fn}^*}{\omega_n + \gamma_f} - \frac{\gamma_m T_{mf}^*}{\omega_n + \gamma_m}$$

$$H_n(\omega_n) = 0$$

# GENERAL REACTOR STABILITY

- ▶ Solving the last equation, we will know the time behavior of the reactor system for given conditions.
- ▶ In fact, the equations as written is of degree  $l + 5$ , where  $l$  is the number of delayed neutron groups considered.
- ▶ There are thus  $(l + 5)$  separate values of  $\omega_n$  (and correspondingly  $(l + 5)$  values of  $A_n, B_n$ , etc.)
- ▶ As mentioned above, of particular interest is the condition that any of these  $(l + 5)$  values of  $\omega_n$  is positive (or has positive real part).
- ▶ For the fundamental mode ( $n = 0$ ) the equation is a generalization of the in-hour equation to take account of feedback effects.

# GENERAL REACTOR STABILITY

- ▶ The general form of this equation is of a rather higher degree than is convenient to handle.

## Further simplifications:

1. Ignore delayed neutrons, or consider only a representative group given by:

$$k_n^* \sum_{i=1}^I \frac{\beta_i}{\omega + \lambda_i} = \frac{\beta}{\omega + \lambda}$$

2. Ignore the Xenon effect in order to study the particular role of the heat-transfer feedback.  
(*This would be appropriate to a core state immediately after start-up.*)

## STABILITY CONDITION

- Returning to  $H_n(\omega_n) = 0$  we may express the equation for  $\omega_n$  in the form:

$$\sum_m B_{nm} \omega_n^m = 0$$

where  $B_{nm}$  are simply  $m$  coefficients obtained from

$$H_n(\omega_n) \equiv l\omega + k_n^* \omega \sum_i \frac{\beta_i}{\omega + \lambda_i} - \alpha_X \phi_n^* \Gamma_n + M^2 \mu_n - \frac{\gamma_f T_{fn}^*}{\omega_n + \gamma_f} - \frac{\gamma_m T_{mf}^*}{\omega_n + \gamma_m}$$

Now when all  $B_{nm}$  are positive there can be no real root, which is greater than zero.

The conditions for complex roots are given by the Routh-Hurwitz criteria, but it may be shown that in our particular case there are no complex roots with positive real part if  $B_{nm} > 0$ .

# STABILITY CONDITION

The condition  $B_{nm} > 0$  for all  $m$  is sufficient to establish the stability of the steady-state reactor to perturbations entering through the  $n^{\text{th}}$  harmonic.

In fact, the critical coefficient turns out to be  $B_{n1}$ , and the stability condition is:

$$M^2 \mu_n - (\alpha_f T_{fn}^* + \alpha_m T_{mn}^*) + \alpha_X (x_n^* - 1 + s) \frac{\phi_n^* \sigma_X}{\lambda_I + \lambda_X + \phi_n^* \sigma_X} > 0$$

The inequality illustrates the fact that the reactor stability is determined by the competition between neutron leakage, heat transfer effects, and  $^{135}\text{Xe}$  fission product poisoning.

# STABILITY CONDITION

- ▶ The **neutron leakage** term is never destabilizing.
  - In the case of the fundamental harmonic ( $n = 0$ ) it is zero, but for higher harmonics the term is stabilizing and increases in magnitude with increasing  $n$ .
- ▶ For the **temperature effect**, it is seen that what is important is the product of the temperature coefficient with the mean steady-state temperature of the material measured above coolant inlet temperature.
  - This temperature is a measure of the efficiency of heat removal from the material.
  - Thus, when  $T^*$  is small, heat is removed efficiently and a change in neutron flux does not lead to appreciable material temperature change.
  - This temperature effect alone is clearly stabilizing provided that:

$$(\alpha_f T_{fn}^* + \alpha_m T_{mn}^*) < 0$$

# STABILITY CONDITION

- ▶ The **Xenon term** is always destabilizing (since  $\alpha_x$  is negative).
  - Its magnitude depends upon the steady-state neutron flux level through the mean reactivity held by  $^{135}\text{Xe}$  and the ratio of the forced to the total decay rate of Xenon and Iodine
  - It is relatively insensitive to the harmonic number  $n$ ; at high flux level the magnitude saturates at the value  $\alpha_x s$ .

# STABILITY CONDITION

- ▶ In a [graphite-moderated reactor](#) only the coefficient  $\alpha_m$  varies significantly with time; this is due to the build-up of Plutonium isotopes in the fuel.
- ▶ It is thus convenient to express the stability criterion in terms of the moderator temperature coefficient of reactivity.
- ▶ The condition for stability is then

$$\alpha_m T_{mn}^* < \frac{M^2}{R_0^2} \mu_n R_0^2 + Q_n - \alpha_f T_{fn}^*$$

where the Xenon term is 
$$Q_n = \frac{\alpha_X (x_n^* - 1 + s) \phi_n^* \sigma_X}{\lambda_I + \lambda_X + \phi_n^* \sigma_X}$$



# SPATIAL INSTABILITIES

- ▶ It may be seen from the stability criterion that, for a given reactor, instability first enters through the fundamental harmonic and then, with increasing  $\alpha_m$ , through higher and higher harmonics.
- ▶ This is basically due to the fact that for a non-fundamental harmonic perturbation neutrons are able to leak or diffuse from the regions of increased neutron density to those regions where the density has fallen below normal.
- ▶ This stabilizing effect becomes more pronounced as the number of such regions of rising and falling neutron density (and hence the harmonic number) increases.
- ▶ The Xenon and temperature feedback terms, on the other hand, are relatively insensitive to the harmonic number.
- ▶ For a given harmonic, the onset of instability is enhanced by increasing power rating (i.e. neutron flux), reactor size, temperature coefficient of reactivity and, for positive coefficients, temperature rise through reactor core.

# TIME CONSTANT OF PERTURBED MOTION

- ▶ The equation for stability condition,  $\sum_m B_{nm} \omega_n^m = 0$ , in principle, enables all the time constants of the perturbed motion to be determined.
- ▶ In general, the time constant of greatest interest (i.e. with largest real part) is complex, i.e.

$$\omega_n = \omega_{n1} + i \omega_{n2}$$

- ▶ The time behavior of the neutron flux perturbation after more rapid transients (smaller values of  $\omega_{n1}$  have decayed) is thus oscillatory and given by:  $\phi' \sim e^{\omega_{n1} t} \cos(\omega_{n2} t + \delta)$

# TIME CONSTANT OF PERTURBED MOTION

- Solutions of small real part are given by an approximate form of the stability condition equation, in which only the first three terms are retained,

$$\sum_{m=0}^2 B_{nm} \omega_n^m = 0$$

and we anticipate that this equation for  $\omega$  is valid in the neighborhood of the stability threshold.

- This quadratic equation yields the general solution (for  $\omega = \omega_1 + i \omega_{n2}$ )

*(next slide)*

# TIME CONSTANT OF PERTURBED MOTION

$$\frac{-2\omega_1}{\lambda_X} = S_n = \frac{L_n \left( 1 + \frac{\lambda_I}{\lambda_X} + \frac{\phi_n^* \sigma_X}{\lambda_X} \right) + \alpha_X \frac{\phi_n^* \sigma_X}{\lambda_X} (x_n^* - 1 + s)}{L_n + \alpha_X \phi_n^* \sigma_X (x_n^* - 1 + s)}$$

$$\frac{4\omega_2^2}{\lambda_X^2} = R_n = \frac{4\lambda_I}{\lambda_X} \left[ S_n - \frac{L_n \lambda_I / \lambda_X + \alpha_X \phi_n^* \sigma_X / \lambda_X}{L_n + \alpha_X \phi_n^* (x_n^* - 1 + s) \tau_m} \right] - S_n^2$$

where 
$$L_n = M^2 \mu_n - (\alpha_f T_{fn}^* + \alpha_m T_{mn}^*)$$

# TIME CONSTANT OF PERTURBED MOTION

- ▶ For a reactor on the threshold of instability ( $\omega_1=0$ ) the motion is oscillatory and the period of oscillation is given by:

$$R_n = \frac{4\lambda_I / \lambda_X}{x_n^* - 1 + s} \left[ \frac{s(1 + \phi_n^* \sigma_X / \lambda_X) + \lambda_I (1 - x_n^*) / \lambda_X}{1 - \lambda_X \tau_m (1 + \lambda_I / \lambda_X + \phi_n^* \sigma_X / \lambda_X)} \right]$$

- ▶ Typically, the period lies in the range 24-30 h and is clearly determined by the xenon-iodine time characteristics.
- ▶ The perturbed motion ceases to be oscillatory when  $R_n \leq 0$ .
- ▶ This will occur whenever  $S_n$  is sufficiently large and negative, i.e. when the reactor is sufficiently over the threshold of instability.
- ▶ In this case the behavior of the reactor is dominated by the non-oscillatory and fairly rapidly-acting temperature effects.

# TEMPERATURE EFFECTS ALONE

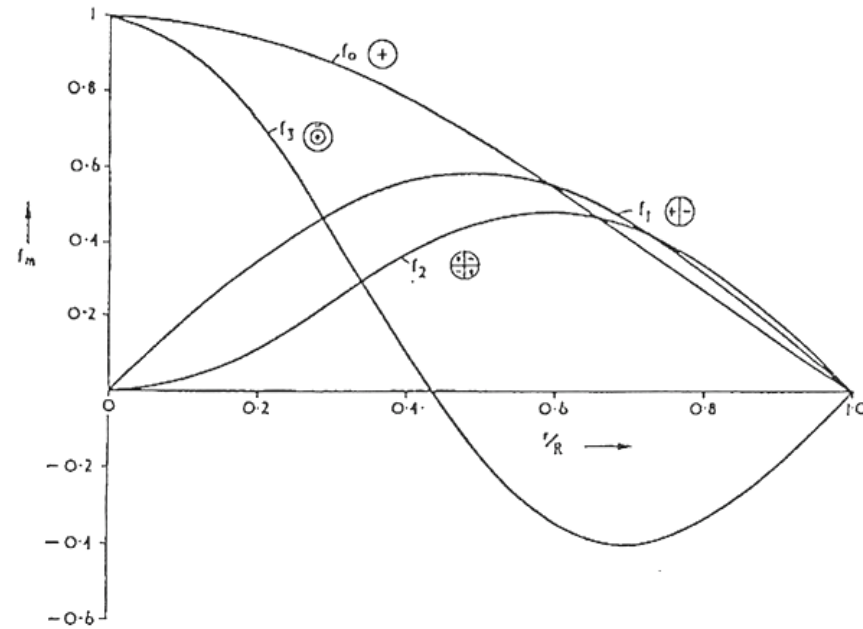
- ▶ For a short while after full power is attained in a reactor there may be little or no  $^{135}\text{Xe}$  present.
- ▶ The state of the reactor is then best described by the foregoing equations without the Xenon terms.
- ▶ In these conditions the condition for stability becomes:

$$M^2 \mu_n - (\alpha_f T_{fn}^* + \alpha_m T_{mn}^*) > 0$$

The critical value of  $\alpha_m T_m^*$  for "temperature only" instability is close to (but slightly greater than) the value of  $\alpha_m T_m^*$  at which in the general case the unstable motion changes from oscillatory to exponential form.

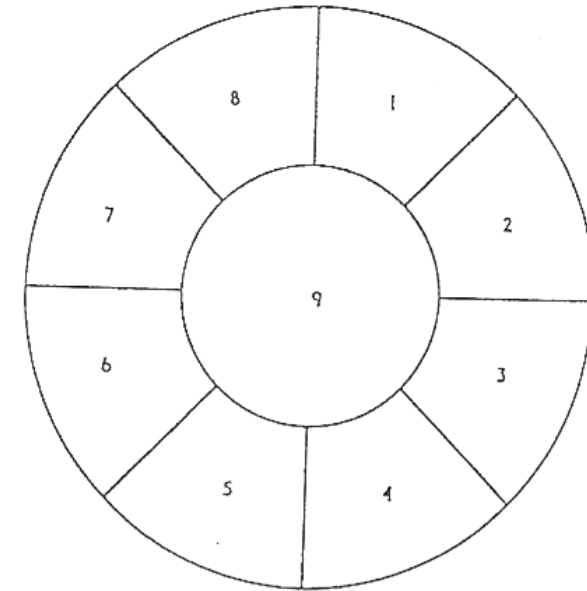
- ▶ Thus, in the general case, the region of purely real time constant may be referred to as the region of "temperature instability".

# CONTROL OF SPATIAL INSTABILITIES



Spatial radial harmonics for a uniform cylindrical reactor. The azimuthal variation is indicated by the circular diagram

To control these harmonics, reactors have to be divided into several sub-regions, with each region controlled independently and the control being applied at the center.



- ▶ With a suitable control system, the harmonic analysis can be extended by including suitable auxiliary equations which describe the control system.
- ▶ Then, the overall stability of the controlled reactor is confirmed, and we can investigate the reactor behavior under assumed faulty controller operation - such as the situation when control in one sub-region is inoperative.

## Next Class:

*Pre-Traditional Multi-Physics Coupling Schemes Thermal-Hydraulics Codes with Point Kinetics Models*